8.6 The Quantum Oscillator

WORKSHEET

The question

For $n = 0, 1, 2, \ldots$, the functions y_n satisfy the differential equation

$$\frac{\mathrm{d}^2 y_n}{\mathrm{d}x^2} - \omega^2 x^2 y_n = -(2n+1)\omega y_n,$$

where ω is a positive constant, and $y_n \to 0$ and $\mathrm{d}y_n/\mathrm{d}x \to 0$ as $x \to +\infty$ and as $x \to -\infty$. Verify that these conditions are satisfied, for $n = 0$ and $n = 1$, by

$$y_0(x) = \mathrm{e}^{-\lambda x^2} \qquad \text{and} \qquad y_1(x) = x\mathrm{e}^{-\lambda x^2}$$

for some constant λ, to be determined.

Show that

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(y_m \frac{\mathrm{d}y_n}{\mathrm{d}x} - y_n \frac{\mathrm{d}y_m}{\mathrm{d}x}\right) = 2(m-n)\omega y_m y_n,$$

and deduce that, if $m \neq n$,

$$\int_{-\infty}^{\infty} y_m(x) y_n(x)\,\mathrm{d}x = 0.$$

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Why should a function like $e^{-\lambda x^2}$ be a sensible guess? What happens to $e^{-\lambda x^2}$ and its derivative as $x \to \pm\infty$?

STAGE 1 The ground state

Ask yourself Differentiate $y_0 = e^{-\lambda x^2}$ twice, substitute into the equation with $n = 0$, and compare coefficients of x^2 and of the constant term.

STAGE 3 An identity

Ask yourself Differentiate $y_m y_n' - y_n y_m'$ by the product rule. Two of the four terms cancel. Then use the differential equation to replace y_n'' and y_m''.

STAGE 4 Orthogonality

Pause and try Integrate the identity from $-\infty$ to ∞. What is the left-hand side?

Try this next

1. **The next state.** Show that $y_2 = (4\lambda x^2 - 1)e^{-\lambda x^2}$, with $\lambda = \frac{\omega}{2}$, satisfies the equation with $n = 2$. Check directly that $\int_{-\infty}^{\infty} y_0 y_2 \, dx = 0$, using $\int_{-\infty}^{\infty} e^{-\omega x^2} \, dx = \sqrt{\pi/\omega}$ and $\int_{-\infty}^{\infty} x^2 e^{-\omega x^2} \, dx = \frac{1}{2\omega}\sqrt{\pi/\omega}$.

2. **No other energies.** Try $y = e^{-\lambda x^2}$ with n replaced by a general number ν. Show that it works only if $\nu = 0$. (In fact, decaying solutions exist only when ν is a non-negative integer.)

3. **Sine waves.** Use the same method to show that $\int_0^{\pi} \sin mx \sin nx \, dx = 0$ for $m \neq n$, starting from $y'' = -n^2 y$ with $y(0) = y(\pi) = 0$.

4. **A ladder.** Show that if y_n solves the equation, then $y_{n+1} = \omega x y_n - y_n'$ solves it with $n + 1$ in place of n. Use this to generate y_1 and y_2 from y_0. (This is the "raising operator" of quantum mechanics.)

Further reading

- **David Griffiths,** ***Introduction to Quantum Mechanics*** (Cambridge University Press). The standard undergraduate physics text; Chapter 2 treats the harmonic oscillator.
- **Richard Feynman,** ***QED: The Strange Theory of Light and Matter*** (Princeton University Press). A popular account of quantum mechanics by one of its masters.
- **Mary Boas,** ***Mathematical Methods in the Physical Sciences*** (Wiley). Covers Hermite polynomials and Sturm–Liouville theory for students of mathematics and physics.

The full guided solution, and what the question leads to at university, is in Section 8.6:
https://beyondstep.keithmcnulty.org/ch08/quantum-oscillator.html